🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.AT.11

Analyze Graphs of Relationships Between Two Variables

Lesson 9-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes
Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use a table and a graph to determine and analyze the relationship between two variable quantities.
Language Objective: I can describe what a point on a graph represents using the words ordered pair, axis, and coordinates.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

graph | table of values | ordered pair | axis | coordinates

👆 *Tap any word to see what it means and a picture.*

1. ________ — A picture on a coordinate plane that shows how two quantities are related.

2. ________ — A list of matched pairs of values that shows a relationship using numbers.

3. ________ — Two numbers written as (x, y) that name one point on a graph. The independent variable comes first.

4. ________ — One of the two number lines that frame a graph. Each axis is labeled with a quantity from the problem.

5. ________ — The numbers in an ordered pair that tell how far to move along each axis to reach a point.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What observations can you make about the data using the graph?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **graph** — A picture on a coordinate plane that shows how two quantities are related.
  gráfica — Un dibujo en un plano de coordenadas que muestra cómo se relacionan dos cantidades.

- [ ] **table of values** — A list of matched pairs of values that shows a relationship using numbers.
  tabla de valores — Una lista de pares de valores que muestra una relación usando números.

- [ ] **ordered pair** — Two numbers written as (x, y) that name one point on a graph. The independent variable comes first.
  par ordenado — Dos números escritos como (x, y) que nombran un punto en la gráfica. La variable independiente va primero.

- [ ] **axis** — One of the two number lines that frame a graph. Each axis is labeled with a quantity from the problem.
  eje — Una de las dos rectas numéricas que enmarcan una gráfica. Cada eje se rotula con una cantidad del problema.

- [ ] **coordinates** — The numbers in an ordered pair that tell how far to move along each axis to reach a point.
  coordenadas — Los números de un par ordenado que indican cuánto avanzar por cada eje para llegar a un punto.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

____ is the independent variable, while ____ is the dependent variable. As the years go by, the interest earned ____, because each year adds $____ to the total.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The number of tickets is the independent variable, and the independent variable always comes first in an ordered pair, so the point is (3, 135), not (135, 3).

Evidence: A strong answer names tickets as independent and explains that independent always goes first, then checks the arithmetic (3 × 45 = 135). A weak answer just says 'because that's the order' without connecting it to independent/dependent.

Reasoning: This evidence connects the definition of graph to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **____ is the independent variable, while ____ is the dependent variable. As the years go by, the interest earned ____, because each year adds $____ to the total.**
- **My answer is ____.**

  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **I know because ____.**

  Lo sé porque ____.
- **So ____.**

  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** graph
2. **Notes 2:** table of values
3. **Notes 3:** ordered pair
4. **Notes 4:** axis
5. **Notes 5:** coordinates
6. **Watch for this mistake:** *Reversing the coordinates of a point — writing (cost, tickets) instead of (tickets, cost). The independent variable always comes first in the ordered pair.*
7. **Try It 1:** A. 6,000 feet *— The remaining distance is 18,000 – 12,000 = 6,000 feet — which matches the last 5 minutes at 1,200 feet per minute.*
8. **Try It 2:** A. The Sandia Peak line (1,200 ft/min) rises more steeply than the 900 ft/min line *— In one minute, the Sandia tram climbs 1,200 feet on the graph while the other climbs only 900 — a faster rate makes a steeper line.*